Mathematical physics made easy(er)

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(Dated: May 15, 2026)

Abstract

Euclidean volume and some distance equations are proved to be instances of ordered combinations (n-tuples) of elements from a list of disjoint, countable sets. A permutation property of volume is proved to limit a set of unordered elements to a cyclic list of at most 3 elements: a set of 3 unordered distance dimensions (height, width, and depth) and a set of 3 unordered non-distance dimensions (time, mass, and charge), each dimension subset R. Where the dimensions of Euclidean distance, time, mass, and charge are each divided into the same number of intervals, there are 3 direct and 3 inverse proportion ratios of the i-th reduced Planck unit interval length of Euclidean distance to the i-th reduced Planck unit interval lengths of time, mass, and charge. Derivations of well-known gravity, charge, electromagnetic, relativity, quantum physics equations and related constants from the ratios are much shorter and simpler. Inverse ratios are used to add quantum extensions to gravity, charge, and relativity equations. The math proofs are verified in Rocq.

[1]

CONTENTS

Keywords: mathematical physics, gravity, charge, electromagnetism, relativity, quantum physics.

INTRODUCTION

Insufficient reduction

The trend in physics has been increasingly complicated mathematics, assumptions, and equations, with increasing difficulty in finding exact solutions to the equations. Examples are Einstein's field equations (general theory of relativity)[2, 3] and Schrödenger's wave equation[4].

And physics equations often rely on assumptions that have not had a mathematical justification. For example, relativity and quantum theory both assume:

1) 3 dimensions of space. But integration, Hilbert spaces, linear algebra, (pseudo-)Riemann manifolds, and so on, have no restriction on the number of dimensions[3, 5–8].

2) The laws of physics are the same at every local Euclidean-like coordinate point[2–4, 9]. But no properties of real and complex numbers have been identified that require functions to be the same at every Euclidean-like coordinate point[5, 6].

Further, much of physics relies on volume and distance equations. Real analysis defines Euclidean volume as the product of interval lengths[5, 6] and defines the Euclidean distance equation, dot product, inner product, and metric space[5, 6]. Justification for the definitions is vigorous finger-pointing to Euclidean and Cartesian geometry.

The philosophy of reductionism posits that complex phenomena can be reduced to a few simple parts: a few simple objects, relations, operations, etc. The first reduction step, in this article, is to prove that well-known geometry equations are reduced to (derived from) a few simple set and number properties. The second step is to identify some consequences of the properties of volume and distance. The third reduction step is to show that well-known physics equations, constants, and assumptions are reduced to (derived from) those consequences.

First reduction step

This section is an overview the first step of proving that well-known geometry equations are derived from a few simple relations between countable set elements.

Each math proof, in this article, has a corresponding formal proof, which has been verified using using the Rocq proof verification system [10], a verification tool used by mathematicians world-wide. The formal proofs are in the Rocq files, "euclidrelations.v" and "threed.v," which are included as ancillary files.

Let $|x_i|$ be the cardinal of (number of elements in) the abstract, countable set, $x_i \in \{x_1, \cdots, x_n\}$. Let the range value, v_c, be the number of ordered combinations (n-tuples) of the elements of $x_1, \cdots, x_n$:

$$x_i \in \{x_1, \cdots, x_n\}, \quad \bigcap_{i=1}^n x_i = \emptyset, \quad |x_i| \in \{0, \mathbb{N}\} \quad \wedge \quad v_c = \prod_{i=1}^n |x_i|. \tag{1}$$

Where each n-tuple corresponds to a real value, $\kappa \in \mathbb{R}$, it will be proved that:

$$v_c = \prod_{i=1}^n |x_i| \quad \Rightarrow \quad v = \prod_{i=1}^n s_i, \quad v, s_i \in \mathbb{R}. \tag{2}$$

For all $n > 1$, there are cases where different domain values, $|x_1|, \cdots, |x_n|$, multiplied yield the same range value, v_c. Inferring a domain value, d_c, from v_c, requires an inverse (bijective) function, $d_c = f_n^{-1}(v_c)$ and $v_c = f_n(d_c)$. The simplest bijective case for all n that includes the $n = 1$ case is the cuboid case, $|x_i| = d_c$. And it will be proved that:

$$v_c = \prod_{i=1}^n |x_i| \quad \wedge \quad d_c = |x_i| \quad \Rightarrow \quad v = d^n. \tag{3}$$

v_c is the sum of the number of subset n-tuples. It will be proved that:

$$d_c^n = v_c = \sum_{i=1}^m v_{ci} \quad \Rightarrow \quad d^n = v = \sum_{i=1}^m (\prod_{j=1}^n s_{i,j}). \tag{4}$$

Note that volume, v, is a scalar quantity and where each $s_{i,j}$ is $\pm$-signed, the $n = 2$ case is the dot product:

$$v = \sum_{i=1}^m (\prod_{j=1}^{n=2} s_{i,j}) \ \wedge \ \exists\, a_i, b_i \in \mathbb{R} : \ s_{i,1} = a_i, \ s_{i,2} = b_i \ \Rightarrow \ v = \sum_{i=1}^m a_i b_i \stackrel{\text{def}}{=} \mathbf{a} \cdot \mathbf{b}. \tag{5}$$

The inner product is where a_i and b_i are also complex numbers, functions, and matrices.

And, where each summed volume, v_{c_i}, is the inverse (bijective) function, $d_{c_i}^n = v_{c_i}$:

$$d_c^n = v_c = \sum_{i=1}^m v_{c_i} = \sum_{i=1}^m d_{c_i}^n \quad \Rightarrow \quad d^n = \sum_{i=1}^m d_i^n. \tag{6}$$

$|d|$ is the p-norm (Minkowski distance)[11], which will be proved to imply the metric space properties[6]. The $n = 2$ case is the Euclidean distance.

Second reduction step

This section provides the rationale leading to the identification of two useful consequences of the set and number properties and corresponding geometry equations.

The commutative property of the union of subsets of n-tuples and the multiplication operation defining the number n-tuples and corresponding Euclidean volume (2) allow n

number of domain elements to be sequenced in all $n!$ permutations, for example, $v = width \times height \times depth = depth \times width \times height = \cdots$. The commutative property implies that each domain element has no intrinsic property that makes an element the first, second, ... , or last, which is a set of unordered elements.

Reliably sequencing a set of unordered elements in the same order requires that the elements be placed into a "list," which assigns relative sequential positions, for example, labeling the unordered elements as $x_1, x_2, \cdots x_n$. The list can be sequenced via the successor and predecessor functions, $successor(x_i) = x_{i+1}$ and $predecessor(x_i) = x_{i-1}$. For example, assigning x_1 = width, x_2 = height, and x_3 = depth, the successor sequence, $v = \prod_{i=1}^{3} s_i$, reliably sequences the unordered set elements in the order, width→height→depth.

Only a cyclic list, where $successor(x_n) = x_1$ and $predecessor(x_1) = x_n$, allows each unordered element to be the first element of some of those $n!$ permutations. For example, where $n = 3$, the volume cyclic successor sequences are: $v = s_1 \times s_2 \times s_3$. $v = s_2 \times s_3 \times s_1$, $v = s_3 \times s_1 \times s_2$, and the cyclic predecessor sequences are: $v = s_1 \times s_3 \times s_2$. $v = s_2 \times s_1 \times s_3$, $v = s_3 \times s_2 \times s_1$.

The only cyclic lists that allows sequencing the unordered elements in all $n!$ permutations, is where each list element is sequentially adjacent (either an *immediate* cyclic successor or an *immediate* cyclic predecessor) to every other element, referred to, in this article, as an immediate symmetric cyclic list (ISCL). For example, the cyclic permutations, $[x_2, x_3, x_1]$ and $[x_2, x_1, x_3]$, require x_1 and x_3 be sequentially adjacent. An ISCL will be proved to contain $n \leq 3$ elements.

1) A consequence of the ISCL property of volume is a set of at most 3 unordered distance dimensions, $\{r_1, r_2, r_3\}$, and a set of at most 3 unordered non-distance dimensions $\{x, y, z\}$, each dimension $\subseteq \mathbb{R}$.

2) A consequence of Euclidean distance being derived as an inverse (bijective) functions of a cuboid volume, $v \in \mathbb{R}$, is the Euclidean distance dimension $\subseteq \mathbb{R}$.

Third reduction step

This section is an overview of the second step of reducing well-known physics equations to the ISCL property of 3 distance distance dimensions and 3 non-distance dimensions, each dimension $\subseteq \mathbb{R}$, and the property of Euclidean distance dimension $\subseteq \mathbb{R}$:

1. **ISCL:** $\{r_1, r_2, r_3\}$ is an ISCL of 3 "distance" dimensions and $\{t\ (time), m\ (mass), q\ (charge)\}$ is the ISCL of 3 "non-distance" dimensions, each dimension $\subseteq \mathbb{R}$. There is a total of 6 unordered dimensions with the $\subseteq \mathbb{R}$ property: $r_1,\ r_2,\ r_3,\ t,\ m,\ q$.

2. **Corresponding lengths:** Where the dimensions of Euclidean distance, time, mass, and charge are each divided into the same number of intervals, the i^{th} unit interval length, r_p of Euclidean distance, corresponds to the i^{th} unit interval lengths: t_p of time, m_p of mass, and q_p of charge, where "unit" means a measurement type, for example, meters, seconds, kilograms, and Coulombs.

The correspondence of unit interval lengths implies 3 direct proportion unit ratios: $r/t = r_p/t_p$, $r/m = r_p/m_p$, and $r/q = r_p/q_p$, letting: $c_t = r_p/t_p$, $c_m = r_p/m_p$, and $c_q = r_p/q_p$. And the correspondence of unit interval lengths implies 3 inverse proportion unit ratios: $rt = t_p r_p$, $rm = m_p r_p$. and $rq = q_p r_p$, And let: $k_t = t_p r_p$, $k_m = m_p r_p$, and $k_q = q_p r_p$.

The gravity[12, 13], special relativity[2, 3], Schwarzschild black hole metric[14, 15], charge[13], electromagnetism[13] equations and related constants are derived from the 3 direct proportion unit ratios. The Planck relation[13], Compton wavelength[13], Schrödenger wave[4], Dirac wave[9] equations and related constants are derived from combining some the direct and inverse proportion unit ratios.

The quantum units: r_p, t_p, m_p, and q_p are derived from the property:

$$r_p = \sqrt{r_p^2} = \sqrt{c_t k_t} = \sqrt{c_m k_m} = \sqrt{c_q k_q}. \tag{7}$$

And it will be shown that r_p, t_p, m_p, and q_p are the reduced Planck units.

The inverse proportion ratios are used to add quantum extensions to Scharwzchild's metric, Newton's gravity, and Coulomb's charge equations. The quantum-extended gravity, charge, and relativity equations predict that where quantum effects are measurable, the constants: gravity, G, Einstein's $\kappa = 8\pi G/c^4$, charge, k_e, permittivity, ε_0, and permeability, μ_0, do not exist. And the quantum-extended equations also predict that as distance between two point masses and between two point charges go to zero, the forces do not go to infinity.

VOLUME

Euclidean volume

Theorem .1. *Euclidean volume,*

$$\forall\, v_c, d_c, |x_i| \in \{0,\ \mathbb{N}\},\ \ x_i \in \{x_1, \cdots, x_n\},\ v_c = \prod_{i=1}^{n} |x_i| \ \Rightarrow\ v = \prod_{i=1}^{n} s_i, \quad s_i, v \in \mathbb{R}. \tag{8}$$

The formal proof, "Euclidean_volume," is in the Rocq file, euclidrelations.v.

Proof.

$$\forall v_c, |x_i| \in \{0, \mathbb{N}\},\ \ \kappa \in \mathbb{R}:\ \ v_c = \prod_{i=1}^{n} |x_i| \ \Leftrightarrow\ v_c\kappa = (\prod_{i=1}^{n} |x_i|)\kappa = \prod_{i=1}^{n}(|x_i|\kappa^{1/n}). \tag{9}$$

$$v_c\kappa = \prod_{i=1}^{n}(|x_i|\kappa^{1/n}) \ \wedge\ \exists\, v, s_i \in \mathbb{R}:\ \ v = v_c\kappa \ \wedge\ s_i = |x_i|\kappa^{1/n} \ \Rightarrow\ v = \prod_{i=1}^{n} s_i. \quad \square \tag{10}$$

Lemma .2. *The number of n-tuples, v_c, is the sum of the number of n-tuples, v_{c_i}, in each disjoint subset of n-tuples, implies a volume is the sum of volumes,*

$$v_c = \sum_{i=1}^{m} v_{c_i} \quad \Rightarrow \quad v = \sum_{i=1}^{m} v_i, \quad v, v_i \in \mathbb{R}.$$

The formal proof, "sum_of _volumes," is in the Rocq file, euclidrelations.v.

Proof. From the condition of this theorem and the Euclidean volume theorem (.1):

$$\forall\, \kappa \in \mathbb{R}:\ \ v_c = \sum_{i=1}^{m} v_{c_i} \quad \Rightarrow \quad v_c\kappa = (\sum_{i=1}^{m} v_{c_i})\kappa = \sum_{i=1}^{m}(v_{c_i}\kappa). \tag{11}$$

$$v_c\kappa = \sum_{i=1}^{m}(v_{c_i}\kappa) \quad \wedge \quad v = v_c\kappa \quad \wedge \quad v_i = v_{c_i}\kappa \quad \Rightarrow \quad v = \sum_{i=1}^{m} v_i. \quad \square \tag{12}$$

DISTANCE

Definition .3. Bijective, countable domain value, d_c:

$$\exists\, v_c, d_c, |x_i| \in \{0,\ \mathbb{N}\},\ \ x_i \in \{x_1, \cdots, x_n\}, \quad v_c = \prod_{i=1}^{n} |x_i| = \prod_{i=1}^{n} d_c = d_c^n. \tag{13}$$

Vector inner product

Theorem .4. *Sum of volumes distance:*

$$d_c^n = v_c = \sum_{i=1}^{m} v_{c_i} \quad \Rightarrow \quad d^n = v = \sum_{i=1}^{m}(\prod_{j=1}^{n} s_{i_j}).$$

The formal proof, "sum_of _volumes _distance," is in the Rocq file, euclidrelations.v.

Proof. Multiply both sides of the assumption by κ and apply the Euclidean volume proof (.1) to v_i:

$$v_c = \sum_{i=1}^{m} v_{c_i} \quad \Rightarrow \quad v = \sum_{i=1}^{m} v_i, \quad v, v_i \in \mathbb{R}. \tag{14}$$

$$v = \sum_{i=1}^{m} v_i \quad \wedge \quad v_i = \prod_{j}^{n} s_{i,j} \quad \Rightarrow \quad v = \sum_{i=1}^{m} (\prod_{j=1}^{n} s_{i,j}). \tag{15}$$

$$d_c^n = v_c \quad \Rightarrow \quad d_c^n \kappa = (d_c \kappa^{1/n})^n = v_c \kappa. \tag{16}$$

$$\exists\, d, v \in \mathbb{R} : d = d_c \kappa^{1/n} \quad \wedge \quad v = v_c \kappa \quad \wedge \quad (d_c \kappa^{1/n})^n = v_c \kappa \quad \Rightarrow \quad d^n = v. \tag{17}$$

$$d^n = v \quad \wedge \quad v = \sum_{i=1}^{m} (\prod_{j=1}^{n} s_{i,j}) \quad \Rightarrow \quad d^n = v = \sum_{i=1}^{m} (\prod_{j=1}^{n} s_{i,j}). \quad \square \tag{18}$$

Volume, v, is a scalar quantity and where each $s_{i,j}$ is $\pm$-signed, the $n = 2$ case is the dot product:

$$v = \sum_{i=1}^{m} (\prod_{j=1}^{n=2} s_{i,j}) \ \wedge\ \exists\, a_{i,1}, a_{i,2} \in \mathbb{R} :\ s_{i,1} = a_i,\ s_{i,2} = b_i \ \Rightarrow\ v = \sum_{i=1}^{m} a_i b_i \stackrel{\text{def}}{=} \mathbf{a} \cdot \mathbf{b}. \tag{19}$$

The inner product is where a_i and b_i are also complex numbers, functions, and matrices.

Minkowski distance (p-norm)

Theorem .5. *Minkowski distance (p-norm):*

$$d_c^n = v_c = \sum_{i=1}^{m} v_{c_i} = \sum_{i=1}^{m} d_{c_i}^n \quad \Leftrightarrow \quad d^n = \sum_{i=1}^{m} d_i^n.$$

The formal proof, "Minkowski_distance," is in the Rocq file, euclidrelations.v.

Proof. From the sum of volumes distance theorem .4:

$$d_{c_i}^n = v_{c_i} \quad \Rightarrow \quad d_i^n = v_i. \tag{20}$$

$$d^n = v = \sum_{i=1}^{m} v_i \quad \wedge \quad d_i^n = v_i \Rightarrow \quad d^n = \sum_{i=1}^{m} d_i^n \quad \square \tag{21}$$

Distance inequalities

Theorem .6. *Distance triangle inequality*

$$\forall\, n \in \mathbb{N},\ \ v_a, v_b \geq 0 :\ \ (v_a + v_b)^{1/n} \leq v_a^{1/n} + v_b^{1/n}.$$

The formal proof, distance_inequality, is in the Rocq file, euclidrelations.v.

Proof. Expand $(v_a^{1/n} + v_b^{1/n})^n$ using the binomial expansion:

$$\forall\, v_a, v_b \geq 0: \quad v_a + v_b \leq v_a + v_b + \textstyle\sum_{i=1}^{n} \binom{n}{k} (v_a^{1/n})^{n-k} (v_b^{1/n})^{k} + \\ \textstyle\sum_{i=1}^{n} \binom{n}{k} (v_a^{1/n})^{k} (v_b^{1/n})^{n-k} = (v_a^{1/n} + v_b^{1/n})^n. \quad (22)$$

Take the n^{th} root of both sides of the inequality 22:

$$\forall\, v_a, v_b \geq 0,\ n \in \mathbb{N}: \quad v_a + v_b \leq (v_a^{1/n} + v_b^{1/n})^n \quad \Rightarrow \quad (v_a + v_b)^{1/n} \leq v_a^{1/n} + v_b^{1/n}. \quad (23)$$

□

Theorem .7. *Distance sum inequality*

$$\forall\, m, n \in \mathbb{N},\ a_i, b_i \geq 0 : (\textstyle\sum_{i=1}^{m} (a_i^n + b_i^n))^{1/n} \leq (\sum_{i=1}^{m} a_i^n)^{1/n} + (\sum_{i=1}^{m} b_i^n)^{1/n}.$$

The formal proof, distance_sum_inequality, is in the Rocq file, euclidrelations.v.

Proof. Apply the distance triangle inequality (.6):

$$\forall\, m, n \in \mathbb{N},\ v_a, v_b \geq 0: \ v_a = \textstyle\sum_{i=1}^{m} a_i^n \quad \wedge \quad v_b = \sum_{i=1}^{m} b_i^n \quad \wedge \quad (v_a + v_b)^{1/n} \leq v_a^{1/n} + v_b^{1/n} \\ \Rightarrow \quad ((\textstyle\sum_{i=1}^{m} a_i^n) + (\sum_{i=1}^{m} b_i^n))^{1/n} = (\sum_{i=1}^{m} (a_i^n + b_i^n))^{1/n} \leq \\ (\textstyle\sum_{i=1}^{m} a_i^n)^{1/n} + (\sum_{i=1}^{m} b_i^n)^{1/n}. \quad \square \quad (24)$$

Metric Space

All Minkowski distances (p-norms) imply the metric space properties. The formal proofs: triangle_inequality, symmetry, non_negativity, and identity_of_indiscernibles are in the Rocq file, euclidrelations.v.

Theorem .8. *Triangle Inequality:*

$$d(s_1, s_2) = (\textstyle\sum_{i=1}^{2} s_i^p)^{1/p}, \quad p \geq 1, \quad u = s_1, \quad w = s_2, \quad v = w/k \\ \Rightarrow \quad d(u, w) = (u^p + w^p)^{1/p} \leq (u^p + v^p)^{1/p} + (v^p + w^p)^{1/p} = d(u, v) + d(v, w).$$

Proof. $\forall\, p \geq 1, \quad u = s_1, \quad w = s_2, \quad v = w/k$:

$$(u^p + w^p)^{1/p} \leq ((u^p + w^p) + 2v^p)^{1/p} = ((u^p + v^p) + (v^p + w^p))^{1/p}. \quad (25)$$

Apply the distance triangle inequality (.6) to the inequality 25:

$$(u^p + w^p)^{1/p} \leq ((u^p + v^p) + (v^p + w^p))^{1/p} \quad \wedge \quad (v_a + v_b)^{1/n} \leq v_a^{1/n} + v_b^{1/n}$$
$$\wedge \quad v_a = u^p + v^p \quad \wedge \quad v_b = v^p + w^p$$
$$\Rightarrow \quad (u^p + w^p)^{1/p} \leq ((u^p + v^p) + (v^p + w^p))^{1/p} \leq (u^p + v^p)^{1/p} + (v^p + w^p)^{1/p}$$
$$\Rightarrow \quad d(u, w) = (u^p + w^p)^{1/p} \leq (u^p + v^p)^{1/p} + (v^p + w^p)^{1/p} = d(u, v) + d(v, w). \quad \square \tag{26}$$

Theorem .9. *Symmetry:*
$d(s_1, s_2) = (\sum_{i=1}^{2} s_i^p)^{1/p}, \quad p \geq 1, \quad u = s_1, \quad v = s_2 \quad \Rightarrow \quad d(u, v) = d(v, u).$

Proof. By the commutative law of addition:

$$\forall\, p \,:\, p \geq 1, \quad d(s_1, s_2) = (\textstyle\sum_{i=1}^{2} s_i^p)^{1/p} = (s_1^p + s_2^p)^{1/p}$$
$$\Rightarrow \quad d(u, v) = (u^p + v^p)^{1/p} = (v^p + u^p)^{1/p} = d(v, u). \quad \square \tag{27}$$

Theorem .10. *Non-negativity:*

$$d(s_1, s_2) = (\textstyle\sum_{i=1}^{2} s_i^p)^{1/p}, \quad p \in N \quad \wedge \quad s_1, s_2 \geq 0 \quad \Rightarrow \quad d(u, v) \geq 0.$$

Proof.

$$\forall\, n \geq 1 \quad \wedge \quad s_1, s_2 \geq 0 \quad \Rightarrow \quad d(s_1, s_2) = (s_1^p + s_1^p)^{1/p} \geq 0. \tag{28}$$
$$\forall\, u = s_1,\ v = s_2 \ \wedge\ d(s_1, s_2) = (s_1^p + s_1^p)^{1/p} \geq 0 \ \Rightarrow\ d(u, v) = (u^p + v^p)^{1/p} \geq 0. \quad \square \tag{29}$$

Theorem .11. *Identity of Indiscernibles:* $d(u, u) = 0$.

Proof. From the non-negativity property (.10):

$$d(u, w) \geq 0 \quad \wedge \quad d(u, v) \geq 0 \quad \wedge \quad d(v, w) \geq 0 \tag{30}$$
$$\Rightarrow \quad \exists\, d(u, w) = d(u, v) = d(v, w) = 0. \tag{31}$$
$$d(u, w) = d(v, w) = 0 \quad \Rightarrow \quad u = v. \tag{32}$$
$$d(u, v) = 0 \quad \wedge \quad u = v \quad \Rightarrow \quad d(u, u) = 0. \quad \square \tag{33}$$

PROPERTIES LIMITING A SET TO AT MOST 3 ELEMENTS

The following definitions and proof use first order logic. A Horn clause-like expression is used, here, to make the proof easier to read. By convention, the proof goal, in a Horn clause, is on the left side of the implication sign ($\leftarrow$) and supporting facts are on the right side. The formal proofs in the Rocq file threed.v are: adj111, adj122, adj212, adj123, adj133, adj233, adj213, adj313, adj323, and not_all_mutually_adjacent_gt_3.

Definition .12. Immediate Cyclic Successor of m is n:

$$\forall\, x_m, x_n \in \{x_1, \cdots, x_{setsize}\}:$$
$$Successor(m, n, setsize) \quad \leftarrow \quad (m = setsize \wedge n = 1) \quad \vee \quad (n = m + 1 \leq setsize). \tag{34}$$

Definition .13. Immediate Cyclic Predecessor of m is n:

$$\forall\, x_m, x_n \in \{x_1, \cdots, x_{setsize}\}:$$
$$Predecessor(m, n, setsize) \quad \leftarrow \quad (m = 1 \wedge n = setsize) \quad \vee \quad (n = m - 1 \geq 1). \tag{35}$$

Definition .14. Adjacent: Element m is sequentially adjacent to element n if the immediate cyclic successor of m is n or the immediate cyclic predecessor of m is n. Notionally:

$$\forall\, x_m, x_n \in \{x_1, \cdots, x_{setsize}\}:$$
$$Adjacent(m, n, setsize) \leftarrow Successor(m, n, setsize) \;\vee\; Predecessor(m, n, setsize). \tag{36}$$

Definition .15. Immediate Symmetric Cylic List (ISCL), where every set element is sequentially adjacent to every other element):

$$\forall\, x_m, x_n \in \{x_1, \cdots, x_{setsize}\}: \quad Adjacent(m, n, setsize). \tag{37}$$

Theorem .16. *An ISCL is limited to at most 3 elements.*

$$\forall\, x_m, x_n \in \{x_1, \cdots, x_{setsize}\}: \quad Adjacent(m, n, setsize) \quad \Rightarrow \quad setsize \leq 3. \tag{38}$$

Proof.

Every element is adjacent to every other element, where $setsize \in \{1,\ 2,\ 3\}$:

$$Adjacent(1,1,1) \leftarrow Successor(1,1,1) \leftarrow (m = setsize \wedge n = 1). \tag{39}$$

$$Adjacent(1,2,2) \leftarrow Successor(1,2,2) \leftarrow (n = m + 1 \leq setsize). \tag{40}$$

$$Adjacent(1,2,3) \leftarrow Successor(1,2,3) \leftarrow (n = m + 1 \leq setsize). \tag{41}$$

$$Adjacent(2,1,3) \leftarrow Predecessor(2,1,3) \leftarrow (n = m - 1 \geq 1). \tag{42}$$

$$Adjacent(3,1,3) \leftarrow Successor(3,1,3) \leftarrow (n = setsize \wedge m = 1). \tag{43}$$

$$Adjacent(1,3,3) \leftarrow Predecessor(1,3,3) \leftarrow (m = 1 \wedge n = setsize). \tag{44}$$

$$Adjacent(2,3,3) \leftarrow Successor(2,3,3) \leftarrow (n = m + 1 \leq setsize). \tag{45}$$

$$Adjacent(3,2,3) \leftarrow Predecessor(3,2,3) \leftarrow (n = m - 1 \geq 1). \tag{46}$$

Element 2 is the only immediate successor of element 1 for all $setsize \geq 3$, which implies element 3 is not ($\neg$) an immediate successor of element 1 for all $setsize \geq 3$:

$$\begin{split}\neg Successor(1,3,setsize \geq 3) \leftarrow \\ Successor(1,2,setsize \geq 3) \leftarrow (n = m + 1 \leq setsize).\end{split} \tag{47}$$

Element $n = setsize > 3$ is the only immediate predecessor of element 1, which implies element 3 is not ($\neg$) an immediate predecessor of element 1 for all $setsize > 3$:

$$\begin{split}\neg Predecessor(1,3,setsize \geq 3) \leftarrow Predecessor(1,setsize,setsize > 3) \leftarrow \\ (m = 1 \wedge n = setsize > 3).\end{split} \tag{48}$$

For all $setsize > 3$, some elements are not ($\neg$) sequentially adjacent to every other element (not immediate symmetric):

$$\begin{split}\neg Adjacent(1,3,setsize > 3) \leftarrow \\ \neg Successor(1,3,setsize > 3) \quad \wedge \quad \neg Predecessor(1,3,setsize > 3). \quad \square\end{split} \tag{49}$$

The Symmetric goal matches the Adjacent goals 39 and fails for all “setsize” greater than three.

DERIVATION OF PHYSICS EQUATIONS AND CONSTANTS

Direct and inverse proportion unit ratios

Application of the ISCL property of volume (.16) and the Euclidean distance dimension $\subseteq \mathbb{R}$ (.5):

1. **ISCL:** $\{r_1, r_2, r_3\}$ is an ISCL of 3 "distance" dimensions and $\{t\ (time), m\ (mass), q\ (charge)\}$ is the ISCL of 3 "non-distance" dimensions, each dimension $\subseteq \mathbb{R}$. There is a total of 6 unordered dimensions with the $\subseteq \mathbb{R}$ property: $r_1,\ r_2,\ r_3,\ t,\ m,\ q$.

2. **Corresponding lengths:** Where the dimensions of Euclidean distance, time, mass, and charge are each divided into the same number of intervals, the i^{th} unit interval length, r_p of Euclidean distance, corresponds to the i^{th} unit interval lengths: t_p of time, m_p of mass, and q_p of charge, where "unit" means a measurement type, for example, meters, seconds, kilograms, and Coulombs.

There are 3 direct proportion ratios:

$$\forall\, r_p, t_p, m_p, q_p \in \mathbb{R},\ \exists\, r, t, m, q \in \mathbb{R} :\ r/t = r_p/t_p \ \wedge\ r/m = r_p/m_p \ \wedge\ r/q = r_p/q_p. \tag{50}$$

$$\exists\, c, c_t, c_m, c_q \in \mathbb{R} : \quad r_p/t_p = c_t \quad \wedge \quad r_p/m_p = c_m \quad \wedge \quad r_p/q_p = c_q$$

$$\Rightarrow \quad r/t = r_p/t_p = c_t = c \quad \wedge \quad r/m = r_p/m_p = c_m \quad \wedge \quad r/q = r_p/q_p. = c_q. \tag{51}$$

And there are 3 inverse proportion ratios:

$$\forall\, r_p, t_p, m_p, q_p \in \mathbb{R},\ \exists\, r, t, m, q \in \mathbb{R} :\ rt = r_p t_p \quad \wedge \quad rm = m_p r_p \quad \wedge \quad rq = q_p r_p. \tag{52}$$

$$\exists\, k_t, k_m, k_q \in \mathbb{R} : \quad r_p t_p = k_t \quad \wedge \quad r_p m_p = k_m \quad \wedge \quad r_p q_p = k_q$$

$$rt = t_p r_p = k_t \quad \wedge \quad rm = m_p r_p = k_m \quad \wedge \quad rq = q_p r_p = k_q. \tag{53}$$

G, Newton, Gauss, Poisson gravity laws

From equations 51, linear acceleration, r/t^2, can be related to mass, m, and distance, r:

$$r = c_m m \ \wedge\ r = c_t t \ \Rightarrow\ r/(c_t t)^2 = c_m m/r^2 \ \Rightarrow\ r/t^2 = (c_m c_t^2) m/r^2 = Gm/r^2, \tag{54}$$

where $G = c_m c_t^2 = r_p^3/m_p t_p^2$, conforms to the SI units: $m^3 \cdot kg^{-1} \cdot s^{-2}$ [12].

Newton's law follows from multiplying both sides of equation 54 by m:

$$r/t^2 = Gm/r^2 \quad \Leftrightarrow \quad F \stackrel{\text{def}}{=} mr/t^2 = Gm^2/r^2. \tag{55}$$

$$F = Gm^2/r^2 \ \wedge \ \forall\, m \in \mathbb{R} : \ \exists\, m_1, m_2 \in \mathbb{R} : \ m_1 m_2 = m^2 \quad \Rightarrow \quad F = Gm_1m_2/r^2. \tag{56}$$

A new interpretation of Gauss's and Poisson's gravity laws is presented here. Equation 54 relates linear acceleration, r/t^2, to mass and distance. Gauss's gravity field, $\mathbf{g}$, and Poisson's gravity field, $-\nabla\Phi(\tilde{\mathbf{r}}, t)$, is the angular (orbital) acceleration, $2\pi r/t^2$. Applying equation 54:

$$r/t^2 = -Gm/|\tilde{\mathbf{r}}|^2 \ \wedge \ \mathbf{g} = -\nabla\Phi(\tilde{\mathbf{r}}, t) = 2\pi r/t^2 \ \Rightarrow \ \mathbf{g} = -\nabla\Phi(\tilde{\mathbf{r}}, t) = -2\pi Gm/|\tilde{\mathbf{r}}|^2. \tag{57}$$

$$\mathbf{g} = -\nabla\Phi(\tilde{\mathbf{r}}, t) = -2\pi Gm/|\tilde{\mathbf{r}}|^2 \quad \Rightarrow \quad \nabla \cdot \mathbf{g} = \nabla^2\Phi(\tilde{\mathbf{r}}, t) = 4\pi Gm/|\tilde{\mathbf{r}}|^3 = 4\pi G\rho, \tag{58}$$

where $\rho = m/|\tilde{\mathbf{r}}|^3$ is the mass density:

Special relativity

$$\forall\, \tau \in \{t, m, q\},\ r^2 = r'^2 + r_v^2,\ \exists\, \mu, \nu : \ r = \mu\tau \quad \wedge \quad r_v = \nu\tau \quad \Rightarrow \quad (\mu\tau)^2 = r'^2 + (\nu\tau)^2$$
$$\Rightarrow \quad r' = \sqrt{(\mu\tau)^2 - (\nu\tau)^2} = \mu\tau\sqrt{1 - (\nu/\mu)^2}. \tag{59}$$

Local frame distance, r', contracts relative to a distant observer frame distance, r, as $\nu \to \mu$:

$$r' = \mu\tau\sqrt{1 - (\nu/\mu)^2} \quad \wedge \quad \mu\tau = r \quad \Rightarrow \quad r' = r\sqrt{1 - (\nu/\mu)^2}. \tag{60}$$

A distant observer frame type, τ, dilates relative to the local observer frame type, τ', as $\nu \to \mu$:

$$\mu\tau = r'/\sqrt{1 - (\nu/\mu)^2} \quad \wedge \quad r' = \mu\tau \quad \Rightarrow \quad \tau = \tau'/\sqrt{1 - (\nu/\mu)^2}. \tag{61}$$

Where τ is type, time, the space-like flat Minkowski spacetime event interval is:

$$\mathrm{d}r^2 = \mathrm{d}r'^2 + \mathrm{d}r_v^2 \quad \wedge \quad \mathrm{d}r_v^2 = \mathrm{d}r_1^2 + \mathrm{d}r_2^2 + \mathrm{d}r_3^2 \quad \wedge \quad \mathrm{d}(\mu\tau) = \mathrm{d}r$$
$$\Rightarrow \ \mathrm{d}r'^2 = \mathrm{d}(\mu\tau)^2 - \mathrm{d}r_1^2 - \mathrm{d}r_2^2 - \mathrm{d}r_3^2. \tag{62}$$

Simple general relativity solutions

The Einstein field equation (EFE) is:

$$G_{\mu,\nu} + \Lambda g_{\mu,\nu} = \kappa T_{\mu,\nu}, \quad G_{\mu,\nu} = \mathbf{R} + \frac{1}{2}Rg_{\mu,\nu}. \tag{63}$$

where: $G_{\mu,\nu}$ is Einstein's tensor, $\mathbf{R}$ is the Ricci curvature, R is the scalar curvature, $g_{\mu,\nu}$ is the metric tensor, Λ is a cosmological constant, $\kappa = 8\pi G/c^4$ is Einstein's constant, and $T_{\mu,\nu}$ is the stress-energy-momentum tensor [2, 3].

The EFE is nonlinear, which has made determining the measure (metric) of the spacetime curvature, $g_{\mu,\nu}$, very difficult. In particular, it has been assumed that the measure of curvature, $g_{\mu,\nu}$, depends on the Ricci curvature, $\mathbf{R}$, and depends on the distribution of the stress, energy, and momentum, $T_{\mu,\nu}$, which bends spacetime [3, 14, 15].

But, in the next subsection, the Schwarzchild metric[14, 15] will be derived independent of the EFE by noting that: 1) The laws of physics are independent of the Ricci curvature and independent of the distribution of the stress, energy, and momentum. 2) The laws of physics are expressed in the components of the metric, $g_{\mu,\nu}$. This leads to the following simple steps to derive all metric tensors, $g_{\mu,\nu}$, independent of the EFE.

Step 1) Use the unit ratios and other equations derived from the ratios, for example, the ratio-derived special relativity equations(60) and the raito-derived constant, G (54), to derive functions (laws) to be assigned to the components of $g_{\nu,\mu}$. For spherically symmetric cases, like the Schwarzchild metric, the ratio derived laws are assigned to the diagonal components of $g_{\mu,\nu}$. For non-symmetric cases, ratio-derived laws are also assigned to the other components of $g_{\mu,\nu}$.

Step 2) The 4D flat spacetime interval equation is an instance of the 2D equation (62). Therefore, derive the metric first as a 2D metric and expand later to a 4D metric.

(Optional step 2d) The 2D metric tensor allows using the much simpler 2D Ricci curvature (which, in 2D, is the Gaussian curvature) and scalar curvature. And the 2D tensors reduce the number of independent equations to solve. The 2D solutions can next be used to set constraints on the solutions in the 4D tensors.

Step 3) Translate from 2D to 4D. For spherically symmetric cases, the simplest method to translate from 2D to 4D is to use spherical coordinates, where the first 2 diagonal components of $g_{\nu,\mu}$ remain unchanged and the other 2 diagonal components are functions of the longitude angle, θ, and latitude angle, ϕ.

Schwarzschild's time dilation and metric

From equations 60 and 51:

$$\sqrt{1-(v^2/c^2)} = \sqrt{1-(1)(v^2/c^2)} \quad \wedge \quad c_m m/r = 1$$
$$\Rightarrow \quad \sqrt{1-(v^2/c^2)} = \sqrt{1-c_m m v^2/rc^2}. \quad (64)$$

Where v_{escape} is the escape velocity:

$$\sqrt{1-(v^2/c^2)} = \sqrt{1-c_m m v^2/rc^2} \quad \wedge \quad KE = mv^2/2 = mv^2_{escape}$$
$$\Rightarrow \quad \sqrt{1-(v^2/c^2)} = \sqrt{1-2c_m m v^2_{escape}/rc^2}. \quad (65)$$

For a photon, the escape velocity, $v_{escape} = c$.

$$\sqrt{1-(v^2/c^2)} = \sqrt{1-2c_m m v^2_{escape}/rc^2}. \quad \wedge \quad v_{escape} = c$$
$$\Rightarrow \quad \sqrt{1-(v^2/c^2)} = \sqrt{1-2c_m mc^2/rc^2}. \quad (66)$$

Combining equation 66 with the derivation of G (54):

$$c_m c^2 = G \wedge \sqrt{1-(v^2/c^2)} = \sqrt{1-2c_m mc^2/rc^2} \Rightarrow \sqrt{1-(v^2/c^2)} = \sqrt{1-2Gm/rc^2}. \quad (67)$$

Combining equation 67 with equation 61 yields Schwarzschild's gravitational time dilation:

$$\sqrt{1-(v^2/c^2)} = \sqrt{1-2Gm/rc^2} \wedge t' = t\sqrt{1-(v^2/c^2)} \Rightarrow t' = t\sqrt{1-2Gm/rc^2}. \quad (68)$$

From equations 60 and 67, where Schwarzschild defined the black hole event horizon radius as $\alpha \stackrel{\text{def}}{=} 2Gm/c^2$ [14, 15]:

$$r' = r\sqrt{1-(v/c)^2} = r\sqrt{1-2Gm/rc^2} \quad \wedge \quad \alpha \stackrel{\text{def}}{=} 2Gm/c^2 \quad \Rightarrow \quad r' = r\sqrt{1-\alpha/r}. \quad (69)$$

Applying equation 69 to the time-like spacetime interval equation 62, where $\mathrm{d}s^2 < 0$:

$$r' = r\sqrt{1-\alpha/r} \quad \wedge \quad \mathrm{d}s^2 = \mathrm{d}r'^2 - \mathrm{d}r^2$$
$$\Rightarrow \quad \mathrm{d}s^2 = (\sqrt{1-\alpha/r}\mathrm{d}r)^2 - (\mathrm{d}r'/\sqrt{1-\alpha/r})^2 = (1-\alpha/r)\mathrm{d}r^2 - (1-\alpha/r)^{-1}\mathrm{d}r'^2. \quad (70)$$

$$\mathrm{d}s^2 = (1-\alpha/r)\mathrm{d}r^2 - (1-\alpha/r)^{-1}\mathrm{d}r'^2 \quad \wedge \quad \mathrm{d}r = \mathrm{d}(ct) \quad \wedge \quad c = 1 \quad \wedge$$
$$\lim_{\mathrm{d}s\to 0} \mathrm{d}r' = \lim_{\mathrm{d}s\to 0} \mathrm{d}r \quad \Rightarrow \quad \lim_{\mathrm{d}s\to 0} \mathrm{d}s^2 = (1-\alpha/r)\mathrm{d}t^2 - (1-\alpha/r)^{-1}\mathrm{d}r^2. \quad (71)$$

Using spherical coordinates to translate from 2D to 4D yields the $+---$ form of Schwarzschild's black hole metric [14, 15]:

$$\begin{aligned}
&\lim_{\mathrm{d}s\to 0} \mathrm{d}s^2 = (1-\alpha/r)\mathrm{d}t^2 - (1-\alpha/r)^{-1}\mathrm{d}r^2 \\
&\quad \Rightarrow \quad \lim_{\mathrm{d}s\to 0} \mathrm{d}s^2 = (1-\alpha/r)\mathrm{d}t^2 - (1-\alpha/r)^{-1}\mathrm{d}r^2 - r^2(\mathrm{d}\theta^2 + sin^2\theta \mathrm{d}\phi^2) \\
&\qquad \Rightarrow \quad g_{\mu,\nu} = diag[(1-\alpha/r),\ -(1-\alpha/r)^{-1},\ -r^2,\ -r^2 sin^2\theta]. \qquad (72)
\end{aligned}$$

k_e, ε_0, E, Coulomb and Gauss laws

From equations 51, linear acceleration, r/t^2, can be related to charge, q, and distance, r:

$$r = c_q q \ \wedge \ r = c_t t \ \Rightarrow \ r/(c_t t)^2 = c_q q/r^2 \ \Rightarrow \ r/t^2 = (c_q c_t^2)q/r^2. \qquad (73)$$

$$r = c_m m \quad \wedge \quad r = c_q q \quad \Rightarrow \quad m = (c_q/c_m)q. \qquad (74)$$

Combining equations 73 and 74 yields Coulombs's law:

$$r/t^2 = (c_q c_t^2)q/r^2 \ \wedge \ m = (c_q/c_m)q \ \Rightarrow \ F \stackrel{\text{def}}{=} mr/t^2 = (c_q^2 c_t^2/c_m)q^2/r^2 = k_e q^2/r^2. \qquad (75)$$

$k_e = c_q^2 c_t^2/c_m = m_p r_p^3/t_p^2 q_p^2$, conforms to the SI units: $kg \cdot m^3 \cdot s^{-2} \cdot C^{-2} \stackrel{\text{def}}{=} N \cdot m^2 \cdot C^{-2}$ [13].

$$\forall\, q \in \mathbb{R}\, \exists\, q_1, q_2 \in \mathbb{R} :\ q_1 q_2 = q^2 \quad \wedge \quad F = k_e q^2/r^2 \quad \Rightarrow \quad F = k_e q_1 q_2/r^2. \qquad (76)$$

A new interpretation of Gauss's electric field law is presented here. Equation 73 relates the linear acceleration, r/t^2, to charge and distance. Gauss's electric field, **E**, relates angular (orbital) acceleration, $2\pi r/t^2$ to charge and distance. Applying equation 75:

$$F_C = mr/t^2 = k_e q^2/r^2 \quad \Rightarrow \quad \exists\, F_E \in \mathbb{R} :\ F_E = 2\pi(mr/t^2) = 2\pi k_e q^2/r^2. \qquad (77)$$

$$F_E = 2\pi k_e q^2/r^2 = q(2\pi(k_e q/r^2)) \quad \wedge \quad \exists\, E \in R :\ E = 2\pi k_e q/r^2 \quad \Rightarrow \quad F_E = qE. \qquad (78)$$

The electric field, $E = 2\pi k_e q/r^2$, conforms to the SI units $kg \cdot m \cdot s^{-2} \cdot C^{-1} \stackrel{\text{def}}{=} N \cdot C^{-1}$.

$$\mathbf{E} = 2\pi k_e q/|\vec{\mathbf{r}}|^2 \quad \Rightarrow \quad \nabla \cdot \mathbf{E} = -4\pi k_e q/|\vec{\mathbf{r}}|^3. \qquad (79)$$

$$\nabla \cdot \mathbf{E} = -4\pi k_e q/|\vec{\mathbf{r}}|^3 \quad \wedge \quad \varepsilon_0 \stackrel{\text{def}}{=} 1/4\pi k_e \quad \wedge \quad \rho \stackrel{\text{def}}{=} q/|\vec{\mathbf{r}}|^3 \quad \Rightarrow \quad \nabla \cdot \mathbf{E} = -\rho/\varepsilon_0, \qquad (80)$$

which is Gauss's electric field law [13] in differential form.

$\mathbf{B}$, μ_0, and Lorentz's law

Applying the distance contraction equation, 60, to equation 78, where r' is the distant observer frame of reference and r is local (rest) frame of reference:

$$r' = r/\sqrt{1 - v^2/c^2} \quad \wedge \quad F' = 2\pi k_e q^2/r'^2 \quad \Rightarrow \quad F = 2\pi k_e q^2(1 - v^2/c^2)/r^2. \tag{81}$$

From equation 78:

$$E = 2\pi k_e q/r^2 \;\wedge\; F = 2\pi k_e q^2(1 - v^2/c^2)/r^2 \;\Rightarrow\; F = q(E - (((2\pi k_e/c^2)q/r^2))v^2). \tag{82}$$

$$F = q(E - (((2\pi k_e/c^2)q/r^2))v^2) \;\wedge\; \exists\, B :\; B = (2\pi k_e/c^2)vq/r^2 \;\Rightarrow\; F = q(E - Bv). \tag{83}$$

Translating equation 83 to vector form:

$$F = q(E - Bv) \quad \Rightarrow \quad \mathbf{F} = q(\mathbf{E} - \mathbf{B} \times \vec{\mathbf{v}}). \tag{84}$$

$$\mathbf{B} \times \vec{\mathbf{v}} = -(\vec{\mathbf{v}} \times \mathbf{B})) \quad \wedge \quad \mathbf{F} = q(\mathbf{E} - \mathbf{B} \times \vec{\mathbf{v}}) \quad \Rightarrow \quad \mathbf{F} = q(\mathbf{E} + \vec{\mathbf{v}} \times \mathbf{B}), \tag{85}$$

which is Lorentz law in the distant observer frame of reference, where the magnetic field, $B = (2\pi k_e/c^2)vq/r^2$, conforms to the base SI units: $kg \cdot s^{-1} \cdot C^{-1} \overset{\text{def}}{=} kg \cdot s^{-2} \cdot A^{-1} \overset{\text{def}}{=} T$.

Based on the assumption (having no math proof-motivation) of a magnetic dipole, the magnetic field divergence outward, $\nabla \cdot \mathbf{B}$, at a point, P, normal to a surface enclosing a charge equals the field divergence inward, $\nabla \cdot \mathbf{B}$:

$$\mathbf{B} = (2\pi k_e/c^2)vq/|\vec{\mathbf{r}}|^2 \quad \Rightarrow \quad \nabla \cdot \mathbf{B} = -(4\pi k_e/c^2)vq/|\vec{\mathbf{r}}|^3 + (4\pi k_e/c^2)vq/|\vec{\mathbf{r}}|^3 = 0. \tag{86}$$

$$\nabla \cdot \mathbf{B} = -(4\pi k_e/c^2)vq/|\vec{\mathbf{r}}|^3 + (4\pi k_e/c^2)vq/|\vec{\mathbf{r}}|^3 = 0 \quad \Rightarrow \quad \mu_0 = (4\pi k_e/c^2), \tag{87}$$

where $\mu_0 = 4\pi k_e/c^2$ conforms to the SI units $kg \cdot m \cdot C^{-2} \overset{\text{def}}{=} kg \cdot m \cdot s^{-2} A^{-2}$.

Note: In the last section of this article, it will be shown that the CODATA definition, $\mu_0 = 2h\alpha/cq_e^2$ [16], and $\mu_0 = 4\pi k_e/c^2$ both reduce to the same equation, $\mu_0 = 4\pi k_m/q_p^2$.

Faraday's law

From the magnetic field equation 84, where the electric and magnetic fields are propagating at the speed, $v = c$:

$$B = (2\pi k_e/c^2)qv/r^2 \quad \wedge \quad v = c \quad \wedge \quad r = ct \quad \Rightarrow \quad B = (2\pi k_e/c^3)q/t^2. \tag{88}$$

$$B = (2\pi k_e/c^3)q/t^2 \quad \Rightarrow \quad \partial B/\partial t = -(4\pi k_e/c^3)q/t^3. \tag{89}$$

$$\partial B/\partial t = -(4\pi k_e/c^3)q/t^3 \quad \wedge \quad r = ct \quad \Rightarrow \quad \partial B/\partial t = -4\pi k_e q/r^3. \tag{90}$$

Faraday's law is the case, where the electric field, $\mathbf{E}$, is a single radial vector, $\mathbf{E_{\tilde{r}_1}}$ acting on some point, P. Therefore, at point, P, $\mathbf{E_{\tilde{r}_2}} = 0$ and $\mathbf{E_{\tilde{r}_3}} = 0$. From the Lorentz equation 85, the magnetic field, $\mathbf{B}$ at point P, is perpendicular to $\mathbf{E_{\tilde{r}_1}}$ (aligned on $\tilde{\mathbf{r}}_2$). For a constant force, the Lorentz equation makes the electric field, $\mathbf{E_{\tilde{r}_1}}$, a function of $\mathbf{B_{\tilde{r}_2}}$. Combining with equation 79:

$$\mathbf{E_{\tilde{r}_1}} = 2\pi k_e q/|\tilde{\mathbf{r}}_2|^2 = f(\mathbf{B_{\tilde{r}_2}}) \quad \wedge \quad \mathbf{E_{\tilde{r}_2}} = 0 \quad \wedge \quad \mathbf{E_{\tilde{r}_3}} = 0 \quad \Rightarrow \quad \nabla \times \mathbf{E} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \frac{\partial}{\partial \tilde{\mathbf{r}}_1^2} & \frac{\partial}{\partial \tilde{\mathbf{r}}_2^2} & \frac{\partial}{\partial \tilde{\mathbf{r}}_3^2} \\ \mathbf{E_{\tilde{r}_1}} & 0 & 0 \end{vmatrix}$$

$$= (\frac{\partial \mathbf{E}_{r_3}}{\partial \tilde{\mathbf{r}}_2^2} - \frac{\partial \mathbf{E}_{r_2}}{\partial \tilde{\mathbf{r}}_3^2})\hat{\mathbf{i}} + (\frac{\partial \mathbf{E}_{r_1}}{\partial \tilde{\mathbf{r}}_3^2} - \frac{\partial \mathbf{E}_{r_3}}{\partial \tilde{\mathbf{r}}_1^2})\hat{\mathbf{j}} + (\frac{\partial \mathbf{E}_{r_2}}{\partial \tilde{\mathbf{r}}_1^2} - \frac{\partial \mathbf{E}_{r_1}}{\partial \tilde{\mathbf{r}}_2^2})\hat{\mathbf{k}}$$

$$= 0\hat{\mathbf{i}} + 0\hat{\mathbf{j}} + (0 - \frac{\partial 2\pi k_e q/|\tilde{\mathbf{r}}_2|^2}{\partial |\tilde{\mathbf{r}}_2|^2})\hat{\mathbf{k}} \quad = \quad 4\pi k_e q/|\vec{\mathbf{r}}_2|^3\hat{\mathbf{k}} \quad \equiv \quad \nabla \times \mathbf{E} = 4\pi k_e q/|\vec{\mathbf{r}}|^3. \quad (91)$$

Combining equations 91 and 90 yields Maxwell-Faraday's law [13]:

$$\nabla \times \mathbf{E} = 4\pi k_e q/|\vec{\mathbf{r}}|^3 \quad \wedge \quad \partial \mathbf{B}/\partial t = -4\pi k_e q/|\vec{\mathbf{r}}|^3 \quad \Rightarrow \quad \nabla \times \mathbf{E} = -\partial \mathbf{B}/\partial t. \quad (92)$$

Maxwell's wave equations

From the Gauss's electric field equation 80:

$$\mathbf{E} = 2\pi k_e q/|\vec{\mathbf{r}}|^2 \quad \wedge \quad r = |\vec{\mathbf{r}}| \quad \Rightarrow \quad \nabla^2 \mathbf{E} = 12\pi k_e q/|\vec{\mathbf{r}}|^4. \quad (93)$$

$$\mathbf{E} = 2\pi k_e q/|\vec{\mathbf{r}}|^2 \ \wedge \ r = ct \quad \Rightarrow \quad \mathbf{E} = 2\pi k_e q/(ct)^2 \quad \Rightarrow \quad \partial^2 \mathbf{E}/\partial t^2 = (12\pi k_e/c^2)q/t^4. \quad (94)$$

$$\partial^2 \mathbf{E}/\partial t^2 = (12\pi k_e/c^2)q/t^4 \quad \wedge \quad r = ct \quad \Rightarrow \quad \partial^2 \mathbf{E}/\partial t^2 = 12\pi k_e c^2 q/|\vec{\mathbf{r}}|^4. \quad (95)$$

$$\nabla^2 \mathbf{E} = 12\pi k_e q/|\vec{\mathbf{r}}|^4 \ \wedge \ \partial^2 \mathbf{E}/\partial t^2 = 12\pi k_e c^2 q/|\vec{\mathbf{r}}|^4 \quad \Rightarrow \quad \nabla^2 \mathbf{E} = (1/c^2)\partial^2 \mathbf{E}/\partial t^2. \quad (96)$$

$$\nabla^2 \mathbf{E} = (1/c^2)\partial^2 \mathbf{E}/\partial t^2 \quad \wedge \quad \varepsilon_0 \mu_0 = 1/c^2 \quad \Rightarrow \quad \nabla^2 \mathbf{E} = \varepsilon_0 \mu_0 \partial^2 \mathbf{E}/\partial t^2, \quad (97)$$

which is Maxwell's electric field wave equation. And from equation (90):

$$\partial B/\partial t = -4\pi k_e q/r^3 \quad \wedge \quad r = |\vec{\mathbf{r}}| \quad \Rightarrow \quad \nabla^2 \mathbf{B} = 12\pi k_e q/|\vec{\mathbf{r}}|^4. \quad (98)$$

$$\mathbf{B} = 2\pi k_e q/|\vec{\mathbf{r}}|^2 \ \wedge \ r = ct \ \Rightarrow \quad \mathbf{B} = 2\pi k_e q/(ct)^2 \ \Rightarrow \quad \partial^2 \mathbf{B}/\partial t^2 = (12\pi k_e/c^2)q/t^4. \quad (99)$$

$$\partial^2 \mathbf{B}/\partial t^2 = (12\pi k_e/c^2)q/t^4 \quad \wedge \quad r = ct \quad \Rightarrow \quad \partial^2 \mathbf{B}/\partial t^2 = 12\pi k_e c^2 q/|\vec{\mathbf{r}}|^4. \quad (100)$$

$$\nabla^2 \mathbf{B} = -12\pi k_e q/|\vec{\mathbf{r}}|^4 \ \wedge \ \partial^2 \mathbf{B}/\partial t^2 = -12\pi k_e c^2 q/|\vec{\mathbf{r}}|^4 \quad \Rightarrow \quad \nabla^2 \mathbf{B} = (1/c^2)\partial^2 \mathbf{E}/\partial t^2. \quad (101)$$

$$\nabla^2 \mathbf{B} = (1/c^2)\partial^2 \mathbf{E}/\partial t^2 \quad \wedge \quad \varepsilon_0 \mu_0 = 1/c^2 \quad \Rightarrow \quad \nabla^2 \mathbf{B} = \varepsilon_0 \mu_0 \partial^2 \mathbf{B}/\partial t^2, \quad (102)$$

which is Maxwell's magnetic field wave equation.

The fine structure constant, α

The ratio of two subtypes of force implies ratios of the form:

$$\alpha_\tau = \frac{F_{\tau_1}}{F_{\tau_2}} = \frac{K\tau_1^2/r^2}{K\tau_2^2/r^2} = \frac{\tau_1^2}{\tau_2^2}. \tag{103}$$

For example, the fine structure electron coupling constant is the ratios of two types of Coulomb's charge force, which simplifies to:

$$\alpha = q_e^2/q_p^2 \approx 0.0072973526, \tag{104}$$

where q_e is the elementary (electron) charge and q_p is the reduced Planck charge unit.

The gravitational fine structure constant, α_G, is the ratio of two subtypes of Newton's gravity force, which simplifies to:

$$\alpha_G = m_e^2/m_p^2, \tag{105}$$

where m_e is the electron mass and m_p is the reduced Planck mass unit.

$\hbar$, h, and Einstein-Planck relation

Applying both the direct proportion ratio (51), and inverse proportion ratio (53):

$$m = k_m/r \quad \wedge \quad r = ct \quad \Rightarrow \quad m(ct)^2 = (k_m/r)r^2 = k_m r. \tag{106}$$

$$m(ct)^2 = k_m r \quad \Rightarrow \quad E \stackrel{\text{def}}{=} mc^2 = k_m r/t^2. \tag{107}$$

$$E = mc^2 = k_m r/t^2 \;\; \wedge \;\; r/t = c \;\; \Rightarrow \;\; E = mc^2 = k_m c/t. \tag{108}$$

The next 4 steps show that $k_m c = \hbar$, the *reduced* Planck constant. From the fine structure equation 105:

$$\alpha = q_e^2/q_p^2 \quad \wedge \quad q_e = 1.60217663 \cdot 10^{-19}\ C \quad \wedge \quad \alpha \approx 7.2973525705 \cdot 10^{-3}$$
$$\Rightarrow \quad q_p = \sqrt{q_e^2/\alpha} \approx 1.87554604 \cdot 10^{-18}\ C. \tag{109}$$

From equations 87 and 109:

$$\mu_0 = 4\pi k_e/c_t^2 = 4\pi c_q^2/c_m = 4\pi(r_p/q_p)^2/(r_p/m_p) = 4\pi m_p r_p/q_p^2 = 4\pi k_m/q_p^2. \tag{110}$$

$$\mu_0 = 4\pi k_m/q_p^2 \;\; \wedge \;\; q_p \approx 1.87554604 \cdot 10^{-18}\ C \;\; \wedge \;\; \mu_0 \approx 1.25663706127(20) \cdot 10^{-6}\ kg\ m\ C^{-2}$$
$$\Rightarrow \quad k_m = \mu_0 q_p^2/4\pi \approx 3.51767294 \cdot 10^{-43}\ kg\ m. \tag{111}$$

$$k_m \approx 3.51767294 \cdot 10^{-43}\ kg\ m \quad \wedge \quad c = 2.99792458 \cdot 10^8\ m\ s^{-1}$$
$$\Rightarrow \quad \hbar = k_m c \approx 1.054571817 \cdot 10^{-34}\ kg\ m^2\ s^{-1}. \tag{112}$$

Combining equations 108 and 112 yields $\hbar$, h. and the Einstein-Planck relation[13, 17]:

$$E = mc^2 = k_m c/t \ kg \ m^2 \ s^{-2} \ \wedge \ k_m c = \hbar \ kg \ m^2 \ s^{-1} \ \Rightarrow \ E = \hbar/t \ kg \ m^2 \ s^{-2}. \tag{113}$$

$$E = \hbar/t \ kg \ m^2 \ s^{-2} \quad \wedge \quad \omega = (2\pi/t) \ radian \ s^{-1} \quad \Rightarrow$$

$$E_\omega = \hbar(2\pi/t) = \hbar\omega \ kg \ m^2 \ radian \ s^{-2}. \tag{114}$$

$$E_\omega = \hbar/t \ kg \ m^2 \ radian \ s^{-2} \quad \wedge$$

$$f = (1/t) \ cycle \ s^{-1} = ((2\pi/t) \ radian \ s^{-1})/(2\pi \ radian \ cycle^{-1})$$

$$\Rightarrow \quad E_f = \hbar\omega(2\pi/2\pi) = 2\pi\hbar f = hf \ kg \ m^2 \ cycle \ s^{-2} \stackrel{\text{def}}{=} hf \ kg \ m^2 \ Hz \ s^{-1}. \tag{115}$$

Unit ratio and reduced Planck unit values

The direct proportion ratios are:

$$c_t = c \approx 2.99792458 \cdot 10^8 \ m \ s^{-1}. \tag{116}$$

$$G = c_m c_t^2 \quad \wedge \quad G \approx 6.67430(15) \cdot 10^{-11} \ m^3 \ kg^{-1} \ s^{-2} \quad \Rightarrow$$

$$c_m = G/c_t^2 \approx 7.426162 \cdot 10^{-28} \ m \ kg^{-1}. \tag{117}$$

$$k_e = c_q^2 c_t^2/c_m \quad \wedge \quad k_e \approx 8.9875517923(13) \cdot 10^9 N m^2 \ C^{-2} \quad \Rightarrow$$

$$c_q = \sqrt{k_e c_m/c_t^2} \approx 8.6175182 \cdot 10^{-18} m \ C^{-1}. \tag{118}$$

From equation 112, the inverse proportion ratio values are:

$$\hbar = k_m c \approx 1.054571783 \cdot 10^{-34} \ kg \ m^2/s \quad \wedge \quad c_t = 2.99792458 \cdot 10^8 \ m/s$$

$$\Rightarrow \quad k_m = \hbar/c_t \approx 3.51767283 \cdot 10^{-43} \ kg \ m. \tag{119}$$

$$r_p = \sqrt{r_p^2} = \sqrt{k_m c_m} \approx 1.616255 \cdot 10^{-35} \ m. \tag{120}$$

$$k_t = r_p/c_t \approx 8.71363 \cdot 10^{-79} \ s \ m. \tag{121}$$

$$k_q = r_p/c_q \approx 3.031361 \cdot 10^{-53} \ C \ m. \tag{122}$$

The reduced Planck unit values are:

$$r_p = \sqrt{c_t k_t} = \sqrt{c_m k_m} = \sqrt{c_q k_q} \approx 1.616255 \cdot 10^{-35} \ m. \tag{123}$$

$$t_p = r_p/c_t \approx 5.391246 \cdot 10^{-44} \ s. \tag{124}$$

$$m_p = r_p/c_m \approx 2.176434 \cdot 10^{-8} \ kg. \tag{125}$$

$$q_p = r_p/c_q \approx 1.875546 \cdot 10^{-18} \ C \quad \textbf{or} \tag{126}$$

$$q_p = \sqrt{q_e^2/\alpha} \approx 1.87554604 \cdot 10^{-18} \ C. \tag{127}$$

Compton wavelength, λ

[13, 17] From equations 53 and 109:

$$\lambda = 2\pi r = 2\pi k_m/m = 2\pi k_m c/mc \quad \wedge \quad h = 2\pi\hbar = 2\pi k_m c \quad \Rightarrow \quad \lambda = h/mc. \tag{128}$$

Schrödenger's wave equation

Start with the previous unit ratio-derived "base" Einstein-Planck relation 113 and multiply the kinetic energy component by mc/mc:

$$mc^2 = \hbar/t \quad \Rightarrow \quad \exists\, V(r,t) :\ \hbar/t = \hbar/2t + V(r,t). \tag{129}$$

$$\hbar/t = h/2t + V(r,t) \quad \Rightarrow \quad \hbar/t = \hbar mc/2mct + V(r,t). \tag{130}$$

And from the distance-to-time (speed of light) ratio (51):

$$\hbar/t = \hbar mc/2mct + V(r,t) \quad \wedge \quad r = ct \quad \Rightarrow \quad \hbar/t = \hbar mc^2/2mcr + V(r,t). \tag{131}$$

$$\hbar/t = \hbar mc^2/2mcr + V(r,t) \ \wedge\ \hbar/t = mc^2 \quad \Rightarrow \quad \hbar/t = \hbar^2/2mcrt + V(r,t). \tag{132}$$

$$\hbar/t = \hbar^2/2mcrt + V(r,t) \quad \wedge \quad r = ct \quad \Rightarrow \quad \hbar/t = \hbar^2/2mr^2 + V(r,t). \tag{133}$$

Multiply both sides of equation 133 by a probability density function, $\Psi(r,t)$.

$$\hbar/t = \hbar^2/2mr^2 + V(r,t) \quad \Rightarrow \quad (\hbar/t)\Psi(r,t) = (\hbar^2/2mr^2)\Psi(r,t) + V(r,t)\Psi(r,t). \tag{134}$$

$$\begin{gathered}(\hbar/t)\Psi(r,t) = ((\hbar)^2/2mr^2)\Psi(r,t) + V(r,t)\Psi(r,t) \quad \wedge \\ \forall\ \Psi(r,t) :\ \partial^2\Psi(r,t)/\partial r^2 = (-1/r^2)\Psi(r,t) \quad \wedge \quad \partial\Psi(r,t)/\partial t = (i\ 1/t)\Psi(r,t) \\ \Rightarrow \quad i\hbar\partial\Psi(r,t)/\partial t = -(\hbar^2/2m)\partial^2\Psi(r,t)/\partial r^2 + V(r,t)\Psi(r,t),\end{gathered} \tag{135}$$

which is the one-dimensional position-space Schrödenger's equation [4, 17].

$$\begin{aligned}&i\hbar\partial\Psi(r,t)/\partial t = -(\hbar^2/2m)\partial^2\Psi(r,t)/\partial r^2 + V(r,t)\Psi(r,t) \quad \wedge \quad ||\vec{\mathbf{r}}|| = r \\ &\qquad \Rightarrow \quad \exists\, \vec{\mathbf{r}} :\ i\hbar\partial\Psi(\vec{\mathbf{r}},t)/\partial t = -(\hbar^2/2m)\partial^2\Psi(\vec{\mathbf{r}},t)/\partial\vec{\mathbf{r}}^2 + V(\vec{\mathbf{r}},t)\Psi(\vec{\mathbf{r}},t),\end{aligned} \tag{136}$$

which is the 3-dimensional position-space Schrödenger's equation [4, 17].

Dirac's wave equation

Start with the previous unit ratio-derived "base" Einstein-Planck relation 113:

$$\hbar/t = mc^2 \quad \Leftrightarrow \quad \hbar/2t + \hbar/2t. = mc^2 \quad \Leftrightarrow \quad \hbar/2t = -\hbar/2t. + mc^2. \tag{137}$$

$$\hbar/2t = -\hbar/2t. + mc^2 \;\wedge\; qc_q/r = qc_q/ct = 1 \;\Rightarrow\; \hbar qc_q/2ct^2 = -qc_q/2rt. + mc^2. \tag{138}$$

$$\hbar qc_q/2ct^2 = -qc_q/2rt. + mc^2 \quad \wedge \quad r = ct \quad \Rightarrow \quad \hbar qc_q/2ct^2 = -qcc_q/2r^2. + mc^2. \tag{139}$$

$$\hbar qc_q/2ct^2 = -qcc_q/2r^2. + mc^2 \quad \Rightarrow \quad \hbar(qc_q/2ct^2)\psi = (-qcc_q/2r^2)\psi + mc^2\psi. \tag{140}$$

$$\psi = e^{-iqc_q((1/2ct)-(1/2r))} \;\Rightarrow\; \partial\psi/\partial t = (-iqc_q/2ct^2)\psi \;\wedge\; \nabla\psi = (iqc_q/2r^2)\psi. \tag{141}$$

$$\begin{aligned} &\partial\psi/\partial t = (-iqc_q/2ct^2)\psi \quad \wedge \quad \nabla\psi = (iqc_q/2r^2)\psi \quad \wedge \\ &\hbar(qc_q/2ct^2)\psi = (-qcc_q/2r^2)\psi + mc^2\psi \quad \Rightarrow \quad i\hbar\partial\psi/\partial t = -i\hbar c\nabla\psi + mc^2\psi. \end{aligned} \tag{142}$$

$$\begin{aligned} &\forall\, y = ax + b \;\exists\, \boldsymbol{\psi}, \boldsymbol{\alpha}, \beta : \; yf(\boldsymbol{\psi}) = \boldsymbol{\alpha} \cdot ax\boldsymbol{\psi} + b\beta\boldsymbol{\psi} \quad \wedge \quad i\hbar\partial\boldsymbol{\psi}/\partial t = -i\hbar c\nabla\boldsymbol{\psi} + mc^2\boldsymbol{\psi} \\ &\Rightarrow \quad \exists\, \boldsymbol{\alpha}, \beta : \; i\hbar\partial\psi/\partial t = i\hbar c\boldsymbol{\alpha} \cdot -\nabla\boldsymbol{\psi} + mc^2\beta\boldsymbol{\psi}, \end{aligned} \tag{143}$$

which is the relativistic, free particle Dirac equation[9], where $\beta = \gamma^0$, $\boldsymbol{\alpha} = \gamma^0\gamma^k$, $k \in \{1, 2, 3\}$, and each γ a matrix.

Total of a type

Applying both the direct (51) and inverse proportion ratios (53) to the Euclidean distance:

$$\begin{aligned} &r = \sqrt{r_1^2 + r_2^2}, \quad r_1 = c_\tau\tau, \quad r_2 = k_\tau/\tau, \quad \tau \in \{t, m, q\} \\ &\Rightarrow \quad r = \sqrt{(c_\tau\tau)^2 + (k_\tau/\tau)^2} \quad \Leftrightarrow \quad \tau = \sqrt{(r/c_\tau)^2 + (k_\tau/r)^2}. \end{aligned} \tag{144}$$

Quantum extension to Schwarzschild's metric

The simplest way to demonstrate how to add quantum physics to general relativity is by extending Schwarzschild's gravitational time dilation equation and black hole metric. Apply the total of a type equation 144 to the derivation of Schwarzschild's time dilation and metric

(64):

$$\sqrt{1-(v^2/c^2)} = \sqrt{1-(v^2/c^2)(r/r)} \quad \wedge \quad r = \sqrt{(c_m m)^2 + (k_m/m)^2} = Q_m$$
$$\Rightarrow \quad \sqrt{1-(v^2/c^2)} = \sqrt{1-Q_m v^2/rc^2}. \tag{145}$$
$$\sqrt{1-(v^2/c^2)} = \sqrt{1-Q_m v^2/rc^2} \quad \wedge \quad KE = mv^2/2 = mv_{escape}^2$$
$$\Rightarrow \quad \sqrt{1-(v^2/c^2)} = \sqrt{1-2Q_m v_{escape}^2/rc^2}. \tag{146}$$

For a photon, the escape velocity, $v_{escape} = c$.

$$\sqrt{1-(v^2/c^2)} = \sqrt{1-2Q_m v_{escape}^2/rc^2}. \wedge\ v_{escape} = c$$
$$\Rightarrow \quad \sqrt{1-(v^2/c^2)} = \sqrt{1-2Q_m c^2/rc^2} = \sqrt{1-2Q_m/r}. \tag{147}$$

Combining equation 147 with equation 61 yields Schwarzschild's gravitational time dilation with a quantum mass effect:

$$\sqrt{1-(v^2/c^2)} = \sqrt{1-2Q_m/r} \quad \wedge \quad t' = t\sqrt{1-(v^2/c^2)} \quad \Rightarrow \quad t' = t\sqrt{1-2Q_m/r}. \tag{148}$$

Schwarzschild defined the black hole event horizon radius, $\alpha \stackrel{\text{def}}{=} 2Gm/c^2$. The radius with the quantum extension is $\alpha \stackrel{\text{def}}{=} 2Q_m$. At this point the exact same equations 69 through 72 yield what looks like the same Schwarzschild black hole metric.

Quantum extension to Newton's gravity force

The quantum mass effect is easier to understand in the context Newton's gravity equation than in general relativity, because the metric equations and solutions in the EFEs are much more complex. From equations 144 and 51:

$$m/\sqrt{(r/c_m)^2 + (k_m/r)^2} = 1 \quad \wedge \quad r^2/(ct)^2 = 1 \quad \Rightarrow \quad r^2/(ct)^2 = m/\sqrt{(r/c_m)^2 + (k_m/r)^2}$$
$$\Rightarrow \quad r^2/t^2 = c^2 m/\sqrt{(r/c_m)^2 + (k_m/r)^2}. \tag{149}$$
$$r^2/t^2 = c^2 m/\sqrt{(r/c_m)^2 + (k_m/r)^2} \quad \Rightarrow \quad (m/r)(r^2/t^2) = (m/r)(c^2 m/\sqrt{(r/c_m)^2 + (k_m/r)^2})$$
$$\Rightarrow \quad F \stackrel{\text{def}}{=} mr/t^2 = c^2 m^2/\sqrt{(r^4/c_m^2) + k_m^2}. \tag{150}$$
$$F = c^2 m^2/\sqrt{(r^4/c_m^2) + k_m^2} \quad \wedge \quad \forall\, m \in \mathbb{R},\ \exists\, m_1, m_2 \in \mathbb{R} :\ m_1 m_2 = m^2$$
$$\Rightarrow \quad F = c^2 m_1 m_2/\sqrt{(r^4/c_m^2) + k_m^2}. \tag{151}$$

Quantum extension to Coulomb's charge force

$$q^2/((r/c_q)^2+(k_q/r)^2)=1 \ \wedge \ r^2/(ct)^2=1 \ \Rightarrow \ r^2/(ct)^2=q^2/((r/c_q)^2+(k_q/r)^2)$$
$$\Rightarrow \quad r^2/t^2=c^2q^2/((r/c_q)^2+(k_q/r)^2). \tag{152}$$

$$r^2/t^2=c^2q^2/((r/c_q)^2+(k_q/r)^2) \quad \wedge \quad r=c_m m$$
$$\Rightarrow \quad F \stackrel{\text{def}}{=} mr/t^2=(c^2/c_m)q^2/((r/c_q)^2+(k_q/r)^2). \tag{153}$$

$$\forall\, q \in \mathbb{R}: \ \exists\, q_1, q_2 \in \mathbb{R}: \ q_1q_2=q^2 \quad \wedge \quad F=(c^2/c_m)q^2/((r/c_q)^2+(k_q/r)^2)$$
$$\Rightarrow \quad F=(c^2/c_m)q^2/((r/c_q)^2+(k_q/r)^2). \tag{154}$$

INSIGHTS AND IMPLICATIONS

1. A distance measure, d, is an inverse (bijective) function of a volume, v, where v is the sum of subset volumes (Euclidean or non-Euclidean), $g(s_{i,1},\cdots,s_{i,n})$. Notionally:

$$\forall\, n \in \mathbb{N}, \ f,g: \ d \geq 0, \ d=f^{-1}(v)=f^{-1}(\textstyle\sum_{i=1}^m g(s_{i,1},\cdots,s_{i,n})): \tag{155}$$

   (a) shows the intimate relation between distance and volume that definitions, like inner product space and metric space, ignore [5, 6];

   (b) is a more simple and concise definition of a distance measure, having the metric space properties [5, 6].

2. The left side of the distance sum inequality (.7),

$$(\textstyle\sum_{i=1}^m(a_i^n+b_i^n))^{1/n} \leq (\sum_{i=1}^m a_i^n)^{1/n}+(\sum_{i=1}^m b_i^n)^{1/n}, \tag{156}$$

   differs from the left side of Minkowski's sum inequality [11]:

$$(\textstyle\sum_{i=1}^m(a_i^n+b_i^n)^{\mathbf{n}})^{1/n} \leq (\sum_{i=1}^m a_i^n)^{1/n}+(\sum_{i=1}^m b_i^n)^{1/n}. \tag{157}$$

   (a) The two inequalities are only the same where $n=1$.

   (b) The distance sum inequality (.7) is a more fundamental inequality because the proof does not require the convexity and Hölder's inequality assumptions required to prove the Minkowski sum inequality [11].

   (c) $\forall\, n>1,\ v>0$: The distance sum inequality term, $v_i^n=a_i^n+b_i^n$: $d=v^{1/n}=(\sum_{i=1}^m v_i^n)^{1/n}$, is the Minkowski distance, which makes the distance sum inequality directly related to geometry. But the Minkowski sum inequality term, $d = v^{1/n} = (\sum_{i=1}^{m}((v_i^n)^{\mathbf{n}}))^{1/n} = (\sum_{i=1}^{m} v_i^{\mathbf{n^2}})^{1/n}$, is *not* a Minkowski distance.

(d) The distance sum inequality might be applicable to machine learning.

3. *Combinatorics.* The number of n-tuples, $v_c = \prod_{i=1}^{n} |x_i|$, was proven to imply the Euclidean volume equation (.1). And the notion of distance was derived as the inverse function of the sum of the number of n-tuples (.4). The Euclidean equation was derived without relying on the geometric primitives and relations in Euclidean geometry [18, 19], axiomatic geometry [20–24], trigonometry [25, 26], calculus [25, 27, 28], and vector analysis [3].

4. *Combinatorics.* The commutative property of the operations defining volume and distance equations requires being able to sequence a set of n number of unordered domain elements in $n!$ number of permuted sequences, which limits the set to $n \leq$ elements (.16). Other elements must have different types (are elements of different sets).

   (a) If there are more than 3 distance and 3 non-distance dimensions $\subseteq \mathbb{R}$, then the additional dimensions are ordered, where each dimension has an intrinsic property having a fixed position in strict totally ordered set.

   (b) If each type of quantum state is an ISCL, then there are at most 3 states of the same type: 3 orientations per dimension of space (+1,0-1), 3 quark color charges, {red, green, blue}, 3 quark anti-color charges, and so on.

5. A discrete (point) valued state has measure 0 (zero-length interval). The ratio of a time, distance, mass, or charge interval length to zero is undefined, which is the reason quantum entangled (discrete) state values exist independent of (non-deterministic with respect to) distance (position), time, mass, and charge. Therefore, a state value in space and time is undetermined until observed.

6. For each unit interval length, r_p, of a 3-dimensional distance, $r \subseteq \mathbb{R}$, there are unit interval lengths of other types of dimensions, forming 3 direct proportion unit ratios (51): $r = (r_p/t_p)t = (r_p/m_p)m = (r_p/q_p)q$ and 3 inverse proportion ratios (53): $r = (r_p t_p)/t = (r_p m_p)/m = (r_p q_p)/q$, where r_p, t_p, m_p, and q_p are the reduced Planck units (123).

7. In this article, the following constants are derived directly from the ratios and reduced Planck constants: gravity, $G = c_m c_t^2 = (r_p/m_p)(r_p/t_p)^2 = r_p^3/m_p t_p^2$ (56), reduced Planck $\hbar = k_m c_t = (r_p m_p)(r_p/t_p) = r_p^2 m_p/t_p$ (109), fine-structure $\alpha = q_e^2/q_p^2$ (105), charge, $k_e = c_q^2 c_t^2/c_m$ (76), vacuum permittivity $\varepsilon_0 = 1/4\pi k_e = 1/4\pi(c_q^2 c_t^2/c_m)$ (80), and vacuum permeability $\mu_0 = 4\pi k_e/c_t^2 = 4\pi c_q^2/c_m = 4\pi r_p m_p/q_p^2$ (86).

   And the quantum extensions to: Schwarzschild's gravitational time dilation (147), Newton's gravity force (151), and Coulomb's charge force (154), show that, where the quantum effects become measurable, the constants G, k_e, ε_0, and μ_0 no longer exist (are no longer valid).

   Therefore, **none** of the following constants: G, $\hbar$, $h = 2\pi\hbar$, α, k_e, ε_0, and μ_0 are fundamental constants because 1) they were all shown to be derived from the ratios and reduced Planck units and 2) some do not exist at all where the quantum effects are measurable.

8. Constants that use notions of temperature and pressure for example, the Boltzmann constant [16], might not be definable solely in terms of unit ratios.

9. Many empirical equations were derived, in this article, using the perspective of mass and charge as dimensions. But there is empirical evidence of mass and charge confined to a point or volume and wave-particles moving through space. The combined direct and inverse ratios and special relativity might be a framework to reconcile the two perspectives and might reduce the pantheon of particles in the Standard Model to a few simple parts.

10. An empirical law *describes* the relations between the variables in an equation. Deriving empirical laws from the ratios *explains* the reasons for relations between the variables in an equation. Further, all the derivations of the physics equations from the ratios are much shorter and simpler than other derivations, which shows that the ratios are an important tool for physicists and engineers.

11. As shown in the subsection deriving the Schwarzschild's gravitational time dilation and black hole metric (64) [14, 15] using ratios illustrates a simple way of finding exact solutions to Einstein's field equations.

12. c_t, c_m, and c_q are the *maximum* ratios: $r = \sqrt{r_1^2 + r_2^2 + r_3^2} \Rightarrow r_1,\ r_2,\ r_3 \le r \Rightarrow \forall\, \tau \in \{t, m, q\} : r_1/\tau,\ r_2/\tau,\ r_3/\tau \le r/\tau = r_p/\tau_p = c_\tau$. For example, the speeds of gravity and light waves are limited to the speed, c_t.

13. c_t is a component of the constants: $G = c_m c_t^2$, $k_e = c_q^2 c_t^2/c_m$, $\varepsilon_0 = 1/4\pi k_e = 1/4\pi(c_q^2 c_t^2/c_m)$, and $\hbar = k_m c_t$. The only constants, derived in this article, that do not contain c_t are: vacuum permeability constant: $\mu_0 = 4\pi k_e/c_t^2 = 4\pi c_q^2/c_m$, and the fine structure constants, $\alpha = q_e^2/q_p^2$ and $\alpha_G = m_e^2/m_p^2$.

14. The fine structure electron coupling constant, α was derived from the ratio of two subtypes of charge force that reduces to the unit-less ratio $\alpha = q_e^2/q_p^2 \approx 0.0072973526$ (105), which is the empirical CODATA value [16].

    (a) The CODATA electron coupling version of the fine structure constant, α, is defined as: $\alpha = q_e^2/4\pi\varepsilon_0\hbar c) = q_e^2/2\varepsilon_0 hc$ [16]. The following steps show that the CODATA definition reduces to the ratio-derived equation:

$$\varepsilon_0 = 1/4\pi k_e = 1/(4\pi(c_q^2 c_t^2/c_m) \quad \wedge \quad \hbar = k_m c_t \quad \wedge \quad h = 2\pi\hbar$$
$$\Rightarrow \quad \varepsilon_0 hc = 2\pi k_m c_t^2/(4\pi(c_q^2/c_m)c_t^2) = k_m/(2(c_q^2/c_m))$$
$$= m_p r_p/(2((r_p/q_p)^2/(r_p/m_p))) = q_p^2/2. \quad (158)$$

$$\alpha = q_e^2/2\varepsilon_0 hc \quad \wedge \quad \varepsilon_0 hc = q_p^2/2 \quad \Rightarrow \quad \alpha = q_e^2/2(q_p^2/2) = q_e^2/q_p^2. \quad (159)$$

    (b) The fine structure electron coupling constant is the ratio of electron static charge force to the propagating quantum charge (photon/electromagnetic) wave force, caused by a moving charged particle.

    (c) The gravitational fine structure electron coupling constant, α_G, can be expressed as the ratio of two subtypes of Newton's gravity force, where m_e is the rest electron mass and m_p is the reduced Planck mass unit: $\alpha_G = m_e^2/m_p^2$ (105).

    The currently used gravitational fine structure constant reduces to the equation derived in this article: $\alpha_G = Gm_e^2/\hbar c = (c_m c_t^2)m_e^2/(k_m c_t)c_t = c_m m_e^2/k_m = (r_p/m_p)m_e^2/m_p r_p = m_e^2/m_p^2$.

15. The assumption that the old definition, $\mu_0 = 4\pi k_e/c_t^2 = 1/\varepsilon_0 c^2$, and the newer NIST/CODATA definition, $\mu_0 = 2h\alpha/cq_e^2$[16], are different because of slightly different values is *incorrect*! The following steps will show that both equations reduce

to the same base equation, $\mu_0 = 4\pi k_m/q_p^2$, which shows that the old and newer NIST/CODATA definitions differ in value because of different measurement methods.

Using the ratios to reduce the old definition of μ_0 derived in this article:

$$\mu_0 = 4\pi k_e/c_t^2 = 4\pi c_q^2/c_m = 4\pi(r_p/q_p)^2/(r_p/m_p) = 4\pi m_p r_p/q_p^2 = 4\pi k_m/q_p^2. \quad (160)$$

Using the ratios to reduce the CODATA definition:

$$h = 2\pi k_m c_t, \;\; \alpha = q_e^q/q_p^2 \;\; \Rightarrow \;\; \mu_0 = 2h\alpha/cq_e^2 = 4\pi k_m c_t(q_e^2/q_p^2)/c_t q_e^2 = 4\pi k_m/q_p^2. \quad (161)$$

16. The derivations of: $\nabla \cdot \mathbf{g} = -4\pi G\rho$ from $\mathbf{g} = 2\pi Gm/|\tilde{\mathbf{r}}|^2$ (57), and $\nabla \cdot \mathbf{E} = -\rho/\varepsilon_0$ from $\mathbf{E} = 2\pi k_e q/|\tilde{\mathbf{r}}|^2$ (80), show that:

    (a) Mass and charge density, ρ, as motivations are unnecessary complexities. Equation 54 used to derive Newton's law and equation 73 used to derive Coulomb's law relate linear acceleration, r/t^2 to mass and charge. It is small step to recognize Gauss's $\mathbf{g}$, Poisson's $-\nabla\Phi(\tilde{\mathbf{r}}, t)$, and Gauss's $\mathbf{E}$ relate the angular (orbital) acceleration, $2\pi r/t^2$, to mass and charge, where differentiation yields the gravity field and electric field laws.

        The derivations are much simpler and shorter using angular (orbital) acceleration, $2\pi r/t^2$. The orbit length, $2\pi r$, corresponds to the Gauss's line integral length of a curve circumscribing a sphere containing the mass or charge.

    (b) Mass and charge density obfuscates the pattern, $\nabla \cdot f(x,r) = -4\pi k_x x/|\tilde{\mathbf{r}}|^3$, being derived from the inverse square pattern, $f(x,r) = 2\pi k_x x/|\tilde{\mathbf{r}}|^2$. The constant, G (derived from the inverse square relation (54)), is used in Einstein's constant, $\kappa = 8\pi G/c^4$, and the use of energy density in the stress energy tensor, $T_{\mu,\nu}$, in Einstein's field equations [3] also obfuscate the inverse square pattern.

17. The ratio-based derivation of Lorentz's law here (84) differs from other relativistic derivations, in that equations for the magnetic field, $\mathbf{B}$, and magnetic permeability, μ_0, are derived in the process. The equation, $\mathbf{B} = (2\pi k_e/c^2)vq/|\vec{\mathbf{r}}|^2$, allows a simple derivation of the Faraday's law (92) and the magnetic component of Maxwell's electromagnetic equations (102).

18. Maxwell's derivation of the electric and magnetic field wave equations required: Gauss's electric field law in a vacuum: $\nabla \cdot \mathbf{E} = -\rho/\varepsilon_0$, Gauss's magnetic field law in a vacuum: $\nabla \cdot \mathbf{B} = 0$, Faraday's Law: $\nabla \times \mathbf{E} = -\partial\mathbf{B}/\partial t$, and Ampere's Law: $\nabla \times \mathbf{B} = \mu_0\varepsilon_0\partial\mathbf{E}/\partial t$[13]. In contrast, the derivations (97) (102), in this article, did not require any of those laws and only required the ratio $r = ct$ and the ratio-derived definitions: $\mathbf{E}$, $\mathbf{B}$, ε_0, and μ_0. And the derivations were shorter and simpler.

19. Einstein's relativity equations: 1) assume the Lorentz transformations, 2) assume the laws of physics are same at each coordinate point, 3) assume the notion of light, and 4) assume that the speed of light is the same at each coordinate point [2, 3].

    The derivations, in this article, were made without those assumptions (does not even require the notion of light). The unit ratios are ratios of Euclidean distance to time, mass, and charge. Therefore, the unit ratios (like the maximum speed ratio, c_t), and all equations (laws) and constants derived from those ratios must be the same at each Euclidean-like coordinate point.

20. The Einstein-Planck relations, $E_\omega = \hbar\omega\ kg\ m^2\ radian\ s^{-2}$ (114) and $E_f = hf\ kg\ m^2\ Hz\ s^{-1}$ (115), are both instances of the ratio-derived base Einstein-Planck relation, $E = \hbar/t\ kg\ m^2\ s^{-2}$ (113). And derivations of Schrödenger's wave equation (136) [4] and Dirac's wave equation (143) [9] from the ratio-derived base Einstein-Planck relation are shorter, simpler, and require fewer assumptions.

21. The quantum extensions to: Schwarzschild's black hole metric (72), Newton's gravity force (151), and Coulomb's charge force (154) make quantifiable predictions:

    (a) For gravity, $\lim_{r\to 0} F = c^2 m_1 m_2/k_m$, and for charge, $\lim_{r\to 0} F = 0$. Finite maximum gravity and charge forces: 1) eliminates the problem of forces going to infinity as $r \to 0$, 2) might allow radioactivity without the need for a weak force, and 3) finite sloped energy well walls might explain quantum tunneling.

    (b) The quantum-extended relativity and classic equations reduce to the non-extended relativity and classic equations and constants, where the distance between 2 masses and between 2 charges is sufficiently large or the masses and charges sufficiently large that the quantum effects are not measurable. *Note* that

G, k_e, ε_0, μ_0, and κ (Einstein's constant, which contains G) do not exist (are not valid), where the quantum effects becomes measurable.

(c) The covariant tensor components, in Einstein's field equations, that had the units $1/distance^2$, will now have the more complex units, $1/\sqrt{(distance^4/c_\tau^2) + k_\tau^2}$, $\tau \in \{t, m\}$.

(d) $1/\sqrt{(distance^4/c_\tau^2) + k_\tau^2}$ implies that as distance $\to 0$, spacetime curvature peaks at the reduced Planck unit distance, r_p (123). The ratio of quantum units, $m_p/r_p^3 \approx 5.154848\dot{1}0^{96}\ kg\ m^{-3}$, might indicate a maximum mass density. A finite force (spacetime curvature) and finite mass density might imply that black holes have sizes > 0 (are not singularities). It might allow black hole evaporation. If there was a "big bang," then it might not have originated from a singularity.

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